

Comptia A+ Assignment

[A+ - Peripherals and Power Supply]

Power Supply Systems

Basics of SMPS and UPS systems. Identify power cables and connectors.

Task

- ❖ **1. List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.**

➤ Devices That Use SMPS and Why It Uses SMPS:

A. Desktop Computer

- A desktop computer uses an SMPS to convert AC power from the wall socket into stable DC voltages required by the motherboard, CPU, RAM, and other internal components.

B. LED Television

- An LED TV uses an SMPS to provide efficient and stable power while reducing heat generation and electricity consumption.

C. Laptop Charger

- A laptop charger contains an SMPS that converts high-voltage AC power into low-voltage DC power required to charge the laptop battery safely.

➤ Devices That Use UPS and Why It Uses UPS:

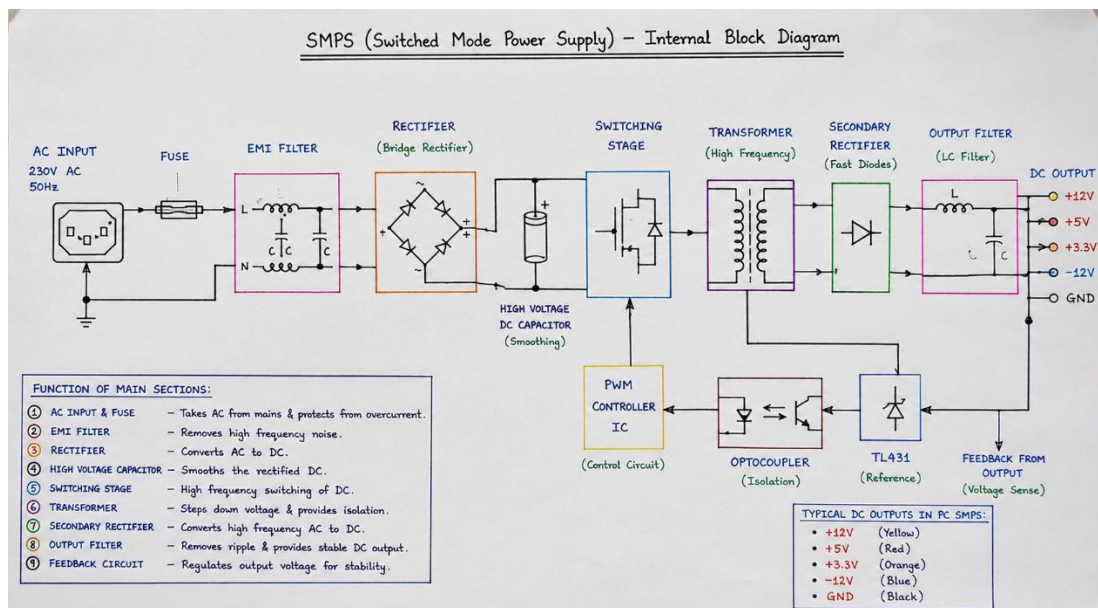
A. Desktop Computer in a Computer Lab

- A UPS provides temporary backup power during a power failure, allowing the user to save work safely and shut down the computer properly.

B. Wi-Fi Router / Networking Equipment

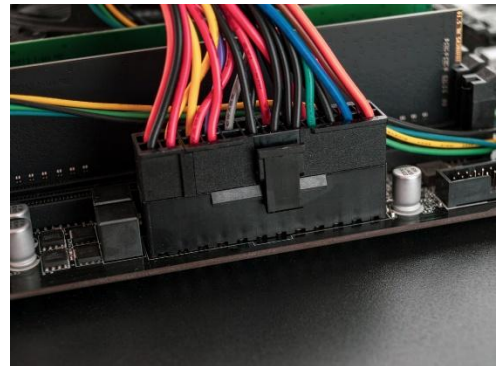
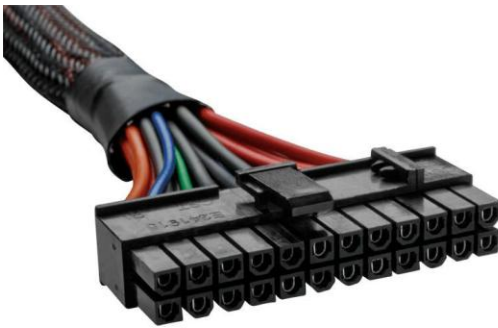
- A UPS keeps the router running during power cuts, ensuring that the internet connection remains available for a limited time.

❖ **2. Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections. You can hand-draw and upload a photo, or use any free diagram tool like draw.**



- ❖ 3. Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.

➤ **ATX 24-Pin Motherboard Connector**



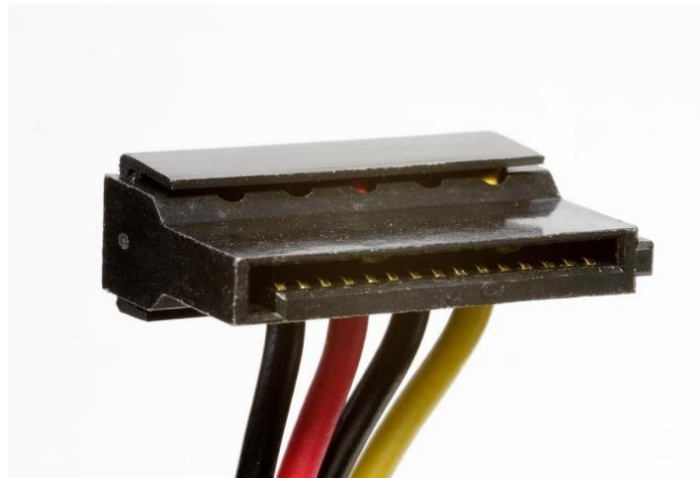
➤ Devices it powers:

- Motherboard
- CPU power distribution
- RAM

➤ Safety Tip:

- Always turn off the computer and unplug the power cable before connecting or disconnecting the 24-pin connector. Press the locking clip before removing it to avoid damaging the motherboard.

➤ **SATA Power Connector**



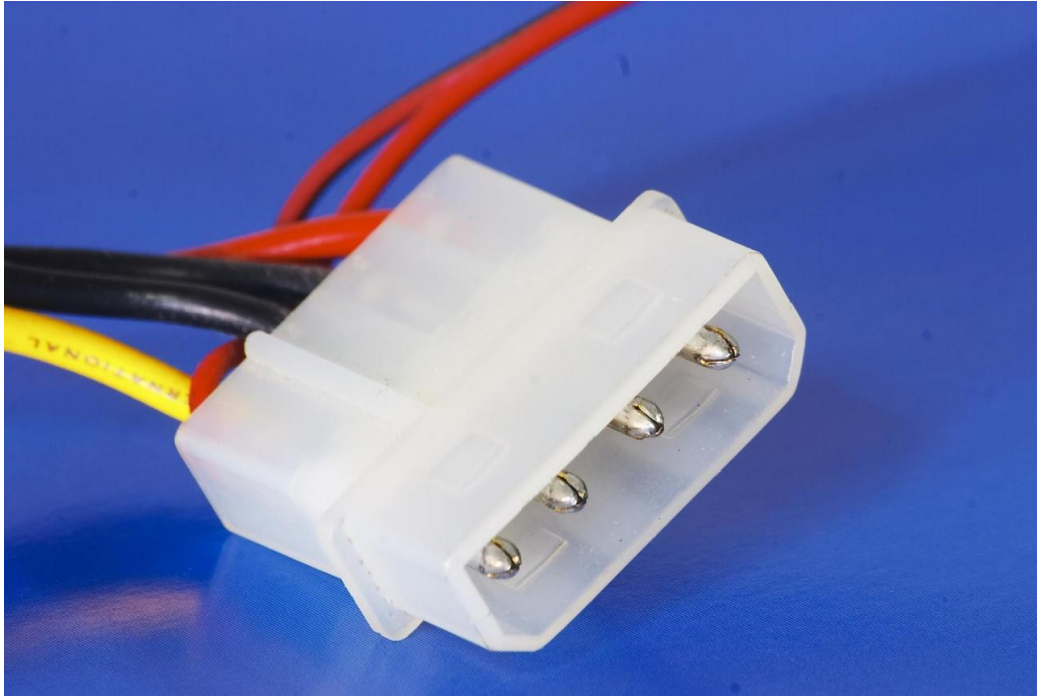
➤ Devices it powers:

- HDD (Hard Disk Drive)
- SSD (Solid State Drive)
- DVD

➤ Safety Tip:

- Do not force the connector into the device. Align the L-shaped notch correctly before inserting or removing it.

➤ **Molex connector**



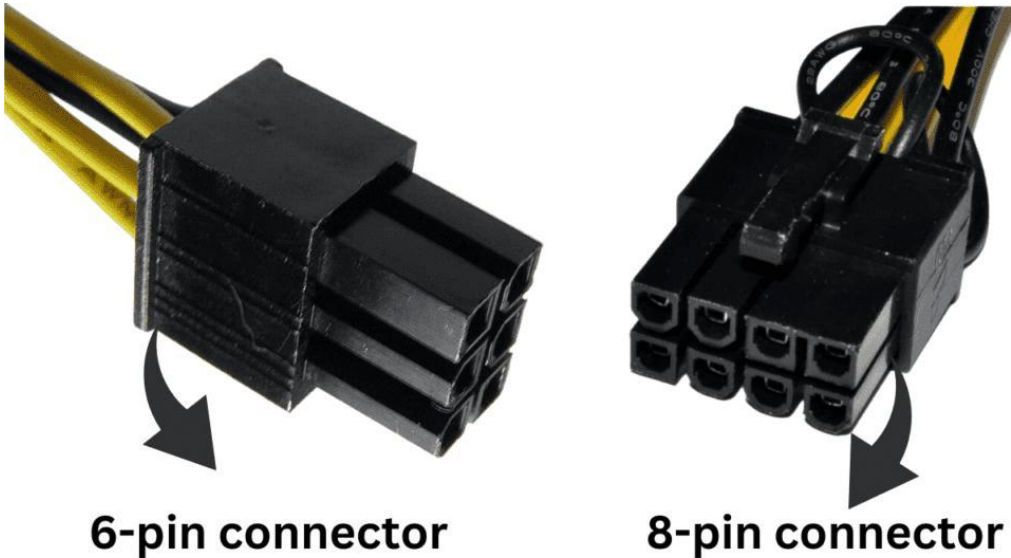
➤ Devices it powers:

- Older IDE hard drives
- Optical drives
- Case fans

➤ Safety Tip:

- Hold the plastic connector while disconnecting it. Never pull on the wires, as this may damage the cable.

➤ **PCIe 6/8-pin connector**



➤ Devices it powers:

- GPU
- High-performance PCIe expansion cards

➤ Safety Tip:

- Ensure the connector is fully locked into the graphics card before powering on the computer. Never connect or disconnect it while the PC is switched on.

❖ **4. Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.**

➤ **Connecting the SMPS in a New Gaming PC**

➤ **Step 1: Install the SMPS**

➤ Description:

- First, I place the SMPS inside the power supply compartment of the computer cabinet and secure it using four screws.

➤ Connector/Cable Used:

- No power cable is connected at this stage (only the SMPS is mounted).

➤ Why This Step Is Important:

- Installing the SMPS first makes it easier to route all the power cables neatly to the components.

➤ **Step 2: Connect the 24-Pin ATX Power Cable to the Motherboard**

➤ Description:

- I connect the 24-pin ATX power connector from the SMPS to the motherboard's main power socket.

➤ Connector/Cable Used:

- **24-Pin ATX Power Connector**

➤ Purpose:

- This cable supplies power to the motherboard, allowing it to distribute electricity to the CPU, RAM, chipset, and other onboard components.

➤ **Step 3: Connect the CPU Power Cable**

➤ Description:

- I connect the CPU power cable near the processor socket on the motherboard.

➤ Connector/Cable Used:

- **8-Pin EPS CPU Power Connector** (or 4+4 Pin)

➤ Purpose:

- This connector provides dedicated power to the processor for stable and reliable performance.

➤ **Step 4: Connect the Graphics Card (GPU)**

➤ Description:

- After installing the graphics card into the PCIe slot, I connect the required power cable from the SMPS.

➤ Connector/Cable Used:

- **PCIe 6-Pin Connector**
- **PCIe 8-Pin Connector** (depending on the graphics card)

➤ Purpose:

- High-performance gaming graphics cards require additional power from the SMPS for smooth graphics processing.

➤ **Step 5: Connect the SSD/HDD**

➤ Description:

- I connect the storage drive to both the motherboard and the SMPS.

➤ Connectors/Cables Used:

- **SATA Power Connector** (from the SMPS)
- **SATA Data Cable** (from the motherboard)

➤ Purpose:

- The SATA power cable provides electrical power, while the SATA data cable allows communication between the storage device and the motherboard.

➤ **Step 7: Verify All Connections**

➤ Description:

- I check that every connector is fully inserted, the locking clips are engaged, and the cables are neatly arranged using cable ties.

➤ Purpose:

- Proper cable management improves airflow, reduces the risk of loose connections, and makes future maintenance easier.

➤ **Step 8: Power On and Test the System**

➤ Description:

- Finally, I connect the AC power cable to the SMPS, switch on the power supply, and press the cabinet power button to test the system.

➤ Purpose:

- This step confirms that the motherboard, graphics card, storage devices, and other components are receiving power and functioning correctly.